

EXPERIMNET 3

1. AIM:

To create a Star Schema data model in Microsoft Power BI using the Global Superstore data.

2. OBJECTIVES:

- To understand the structure of a Star Schema.
- To identify suitable fact and dimension attributes from business data.
- To organize transactional data into a dimensional model.
- To create separate dimension tables for analysis.
- To connect dimension tables with a central fact table.
- To use DAX measures for analyzing business performance.
- To verify the accuracy of the developed data model.

• TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor – built into Power BI Desktop
- DAX (Data Analysis Expressions)

3. DATASET USED: Global Superstore Dataset.xlsx.

4. THEORY:

Dimensional Modeling is a method of organizing data into structures that make business analysis and reporting easier. It mainly consists of **Fact Tables** and **Dimension Tables**.

Fact Table stores business transactions and measurable values such as Sales, Profit, Quantity, and Discount. It generally forms the center of the analytical model.

Dimension Tables contain descriptive information used to analyze the facts. Examples include Customer, Product, Region, and Date.

Star Schema is a dimensional model where a central fact table is directly connected to multiple dimension tables. This structure makes the data model simple, organized, and easier to use for reporting.

In this experiment, the Global Superstore **Orders** data is organized into a central Fact_Sales table. Separate tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date provide descriptive information for analysis.

5. PROCEDURE:

Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.

Launch Microsoft Power BI Desktop.

Step 2: Import the Global Superstore dataset.

Home > Get Data > Excel Workbook

Select: Global Superstore Dataset.xlsx

Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:

The screenshot shows the Microsoft Power BI Desktop interface. On the left, the 'Navigator' pane displays the 'Global Superstore 2018.xlsx' file with three worksheets selected: 'Orders', 'People', and 'Returns'. The main view shows the 'Returns' table with the following data:

Column1	Column2	Column3
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia
Yes	IN-2017-TS212051-42904	Southern Asia
Yes	AG-2014-RO97803-41695	North Africa
Yes	AG-2015-LC70503-42265	North Africa
Yes	AG-2014-DB32103-41987	North Africa
Yes	AG-2017-AC4203-42796	North Africa
Yes	AG-2017-RD96603-43073	North Africa
Yes	AG-2015-CP23403-42152	North Africa
Yes	AG-2016-HF49953-42437	North Africa
Yes	AG-2015-CC24303-42247	North Africa
Yes	AG-2017-MO78003-42948	North Africa
Yes	AO-2017-BE14554-42954	Central Africa
Yes	AO-2017-GM45004-42996	Central Africa
Yes	AO-2016-IG50854-42668	Central Africa
Yes	US-2017-SR204255-42937	South America
Yes	US-2014-JP155205-41883	South America
Yes	MX-2016-LC171405-42457	South America
Yes	US-2016-JK152055-42585	South America
Yes	US-2014-IC153405-41897	South America
Yes	US-2017-CA119655-43061	South America
Yes	US-2016-AB1109455-42505	South America

At the bottom of the main view, there are buttons for 'Load', 'Transform Data', and 'Cancel'. The 'Load' button is highlighted in green.

- Orders
- Returns
- People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

Step 4: Analyse the Orders table.

Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity

- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.

Table: Orders (51,290 rows)

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Co
27884	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
27883	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
26896	ID-2014-FP143207-41945	Sunday, November 2, 2014	Saturday, November 8, 2014	Standard Class	FP-143207	Frank Preis	Consumer	
30039	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30040	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30041	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
27444	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27443	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27445	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27446	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
25656	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25655	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25657	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25658	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
27561	ID-2015-CS124607-42189	Saturday, July 4, 2015	Wednesday, July 8, 2015	Standard Class	CS-124607	Chuck Sachs	Consumer	
28358	IN-2016-JF154907-42399	Saturday, January 30, 2016	Thursday, February 4, 2016	Standard Class	JF-154907	Jeremy Farry	Consumer	
24819	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
24818	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
22411	ID-2016-DJ136307-42524	Friday, June 3, 2016	Thursday, June 9, 2016	Standard Class	DJ-136307	Doug Jacobs	Consumer	
22118	IN-2016-LS172307-42490	Saturday, April 30, 2016	Friday, May 6, 2016	Standard Class	LS-172307	Lycoris Saunders	Consumer	

Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data.

Organize the attributes into:

A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

This classification forms the basis of the Star Schema.

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.

Home > Transform Data

Step 8: Create Dim_Customer.

The screenshot shows the Power Query Editor with a query named 'Dim_Customer'. The formula bar displays the M code: `= Table.TransformColumns(#"Removed Blank Rows",{"Customer ID", Text.Proper, type text}, {"Customer`. The data table has three columns: Customer ID, Customer Name, and Segment. The right-hand pane shows the 'Query Settings' for 'Dim_Customer', with 'Capitalized Each Word' selected under 'APPLIED STEPS'.

Customer ID	Customer Name	Segment
Ca-120551	Cathy Armstrong	Home Office
Bd-116051	Brian Dahlen	Consumer
Aj-107801	Anthony Jacobs	Corporate
Gm-144551	Gary Mitchum	Home Office
Vb-217451	Victoria Brennan	Corporate
Lo-171701	Lori Olson	Corporate
Ss-201401	Saphira Shifley	Corporate
Aa-103751	Allen Arnold	Consumer
Bg-110351	Barry Gonzalez	Consumer
Ah-105851	Angele Hood	Consumer
Cs-118451	Cari Sayre	Corporate
Dw-131951	David Wiener	Corporate
Rs-194201	Ricardo Sperren	Corporate
Ts-213401	Toby Swindell	Consumer
Jg-151151	Jack Garza	Consumer
Su-206651	Stephanie Ulpright	Home Office
Di-134951	Dionis Lloyd	Corporate
Eh-139451	Eric Hoffmann	Consumer
Ed-138851	Emily Ducich	Home Office
Jm-156551	Jim Mitchum	Corporate
Ts-212051	Thomas Seio	Corporate

Create a customer dimension containing:

- Customer ID
- Customer Name
- Segment

Remove duplicate records from the dimension.

Step 9: Create Dim_Product.

The screenshot shows the Power Query Editor with a query named 'Dim_Product'. The formula bar displays the M code: `= Table.TransformColumns(#"Removed Blank Rows",{"Product ID", Text.Proper, type text}, {"Product`. The data table has four columns: Product ID, Product Name, Sub-Category, and Category. The right-hand pane shows the 'Query Settings' for 'Dim_Product', with 'Capitalized Each Word' selected under 'APPLIED STEPS'.

Product ID	Product Name	Sub-Category	Category
1 861	Ikea Library With Doors, Mobile	Bookcases	Furniture
2 988	Acme Scissors, Easy Grip	Supplies	Office Supplies
3 1211	Epson Receipt Printer, White	Machines	Technology
4 726	Rubbermaid Door Stop, Ergonomic	Furnishings	Furniture
5 564	Cameo Interoffice Envelope, With Clear Poly Window	Envelopes	Office Supplies
6 420	Bevis Conference Table, Fully Assembled	Tables	Furniture
7 626	Bush Classic Bookcase, Pine	Bookcases	Furniture
8 583	Hon Rocking Chair, Red	Chairs	Furniture
9 814	Samsung Audio Dock, Voip	Phones	Technology
10 129	Apple Audio Dock, Voip	Phones	Technology
11 780	Chromcraft Wood Table, With Bottom Storage	Tables	Furniture
12 747	Hoover Toaster, Red	Appliances	Office Supplies
13 588	Hewlett Personal Copier, High-Speed	Copiers	Technology
14 523	Harbour Creations Chairmat, Black	Chairs	Furniture
15 68	Smead Trays, Wire Frame	Storage	Office Supplies
16 758	Chromcraft Coffee Table, With Bottom Storage	Tables	Furniture
17 251	Motorola Headset, Cordless	Phones	Technology
18 08	Rogers Shelving, Wire Frame	Storage	Office Supplies
19 727	Rubbermaid Frame, Black	Furnishings	Furniture
20 812	Samsung Audio Dock, Cordless	Phones	Technology
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Create a product dimension containing:

- Product ID
- Product Name
- Category
- Sub-Category

Remove duplicate records.

Step 10: Create Dim_Region.

Create a region dimension containing:

The screenshot shows the Power Query Editor interface. The main area displays a table with the following data:

	City	State	Country	Region
1	Herat	Hirat	Afghanistan	Southern Asia
2	Kabul	Kabul	Afghanistan	Southern Asia
3	Kandahar	Kandahar	Afghanistan	Southern Asia
4	Jalalabad	Nangarhar	Afghanistan	Southern Asia
5	Durres	Durrës	Albania	Southern Europe
6	Elbasan	Elbasan	Albania	Southern Europe
7	Korce	Korçë	Albania	Southern Europe
8	Shkoder	Shkodër	Albania	Southern Europe
9	Vlore	Vlorë	Albania	Southern Europe
10	Algiers	Alger	Algeria	North Africa
11	Annaba	Annaba	Algeria	North Africa
12	Barika	Batna	Algeria	North Africa
13	Batna	Batna	Algeria	North Africa
14	Bechar	Bechar	Algeria	North Africa
15	Bejaia	Bejaia	Algeria	North Africa
16	Constantine	Constantine	Algeria	North Africa
17	Ain Oussera	Djelfa	Algeria	North Africa
18	Dar Chioukh	Djelfa	Algeria	North Africa
19	Messaad	Djelfa	Algeria	North Africa
20	Guelma	Guelma	Algeria	North Africa
21	Laghouat	Laghouat	Algeria	North Africa

The interface includes a ribbon with tabs: File, Home, Transform, Add Column, View, Tools, Help. The 'Transform' tab is active, showing options like 'Remove Duplicates', 'Sort', 'Split Column', 'Group By', 'Replace Values', 'Merge Queries', 'Append Queries', and 'Combine Files'. The 'Query Settings' pane on the right shows the 'Name' as 'Dim_Region' and the 'Applied Steps' list including 'Source', 'Removed Other Columns', 'Capitalized Each Word', 'Removed Duplicates', and 'Removed Blank Rows'.

- Region
- Country
- State
- City

Remove duplicate records.

Step 11: Create Dim_Date.

Create a date dimension containing:

- Order Date
- Year
- Quarter
- Month

Ensure that the date-related fields are correctly formatted.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files

Queries [7]

fx = Table.SelectRows("#Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))

	Order Date	Year	Quarter	Month Name
1	3/22/2017	2017	1	March
2	9/1/2015	2015	3	September
3	4/19/2015	2015	2	April
4	8/1/2017	2017	3	August
5	12/11/2017	2017	4	December
6	9/24/2016	2016	3	September
7	12/16/2015	2015	4	December
8	7/10/2017	2017	3	July
9	9/28/2015	2015	3	September
10	11/27/2016	2016	4	November
11	1/18/2016	2016	1	January
12	11/30/2014	2014	4	November
13	6/5/2015	2015	2	June
14	8/16/2014	2014	3	August
15	12/23/2017	2017	4	December
16	10/24/2017	2017	4	October
17	9/17/2016	2016	3	September
18	7/29/2014	2014	3	July
19	9/2/2017	2017	3	September
20	12/12/2014	2014	4	December
21	9/23/2016	2016	3	September

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 3:55 PM

Step 12: Remove duplicate records.

Remove duplicate records from all dimension tables and verify that the key fields contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows Transpose Reverse Rows Detect Data Type Fill Pivot Column Unpivot Columns Merge Columns Split Column Format Text Column Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration Run R script Run Python script

Queries [8]

fx = Table.TransformColumns(Table.TransformColumnTypes("#Removed Blank Rows", {"Order Date", type text}, {"Sales", type text}, {"Quantity",

	Order ID	Order Date	Customer ID	Region	Product ID	Sales	Quantity
1	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Fur-Bo-4861	731.82	2
2	Id-2015-Bd116051-42248	9/1/2015	Bd-116051	Southern Asia	Off-Su-2988	243.54	9
3	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Tec-Ma-4211	346.32	3
4	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Fur-Fu-5726	169.68	4
5	Id-2015-Bd116051-42248	9/1/2015	Bd-116051	Southern Asia	Off-En-3664	203.88	4
6	Id-2015-Aj107801-42113	4/19/2015	Aj-107801	Southern Asia	Fur-Ta-3420	4626.1499999999996	5
7	In-2017-Gm144551-42948	8/1/2017	Gm-144551	Southern Asia	Fur-Bo-3626	2070.15	5
8	In-2017-Vb217451-43080	12/11/2017	Vb-217451	Southern Asia	Fur-Ch-4683	914.34	7
9	In-2016-Lo171701-42637	9/24/2016	Lo-171701	Southern Asia	Tec-Ph-5814	1168.44	7
10	In-2017-Vb217451-43080	12/11/2017	Vb-217451	Southern Asia	Tec-Ph-3129	500.94	3
11	Id-2015-Ss201401-42354	12/16/2015	Ss-201401	Southern Asia	Fur-Ta-3780	1447.56	3
12	In-2017-Aa103751-42926	7/10/2017	Aa-103751	Southern Asia	Off-Ap-4747	669.12	8
13	In-2015-Bg110351-42275	9/28/2015	Bg-110351	Southern Asia	Tec-Co-4588	427.41	3
14	In-2016-Ah105851-42701	11/27/2016	Ah-105851	Southern Asia	Fur-Ch-4523	417.42	6
15	In-2016-Cs118451-42387	1/18/2016	Cs-118451	Southern Asia	Off-St-6068	332.85	7
16	In-2016-Cs118451-42387	1/18/2016	Cs-118451	Southern Asia	Fur-Ta-3758	267.08999999999997	1
17	In-2015-Bg110351-42275	9/28/2015	Bg-110351	Southern Asia	Tec-Ph-5251	412.95	5
18	In-2017-Aa103751-42926	7/10/2017	Aa-103751	Southern Asia	Off-St-5708	244.8	4
19	In-2014-Ah105851-41973	11/30/2014	Ah-105851	Southern Asia	Fur-Fu-5727	219.78	2
20	In-2015-Dw131951-42160	6/5/2015	Dw-131951	Southern Asia	Tec-Ph-5812	848.4	5
21							

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 3:55 PM

Create the central Fact Table containing:

- Order ID
- Customer ID
- Product ID
- Region
- Order Date
- Sales
- Quantity
- Profit
- Discount

Step 14: Verify key columns.

Verify that all key columns required to establish relationships with the Dimension Tables are present and correctly formatted.

The resulting Fact_Sales table acts as the central table of the Star Schema.

Part E: Build Star Schema

Step 15: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.

Step 16: Create relationships.

Create relationships between:

The screenshot shows the 'Manage relationships' window in Power BI. The window has a title bar with 'EXP-3 & EXP-4 PBI'. Below the title bar is a ribbon with 'File', 'Home', and 'Help'. The main area is titled 'Manage relationships' and contains a table of relationships. The table has columns: 'From: table (column)', 'Relationship', 'To: table (column)', and 'Status'. There are four rows of relationships, all with a status of 'Active'. The relationships are: Fact_Sales (Customer ID) to Dim_Customers (Customer ID), Fact_Sales (Order Date) to Dim_Date (Order Date), Fact_Sales (Product ID) to Dim_Products (Product ID), and Fact_Sales (Region) to Dim_Region (Region). The 'Fact_Sales (Region)' relationship is a many-to-many relationship, while the others are one-to-many. The 'Fact_Sales' table is the central fact table, and the dimension tables are Dim_Customers, Dim_Date, Dim_Products, and Dim_Region. The 'Fact_Sales' table has 17,415 rows.

From: table (column)	Relationship	To: table (column)	Status
Fact_Sales (Customer ID)	* — 1	Dim_Customers (Customer ID)	Active
Fact_Sales (Order Date)	* — 1	Dim_Date (Order Date)	Active
Fact_Sales (Product ID)	* — 1	Dim_Products (Product ID)	Active
Fact_Sales (Region)	* — *	Dim_Region (Region)	Active

Table: Dim_Customers (17,415 rows)

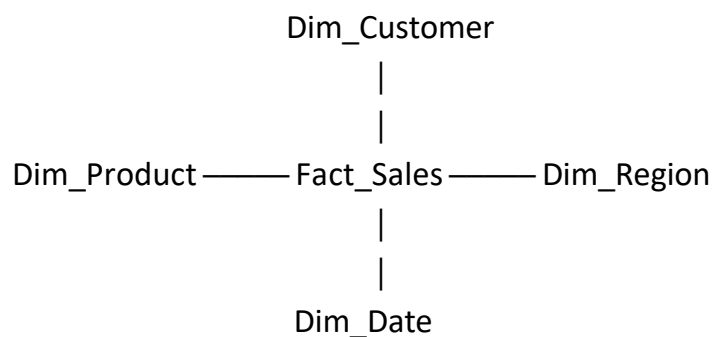
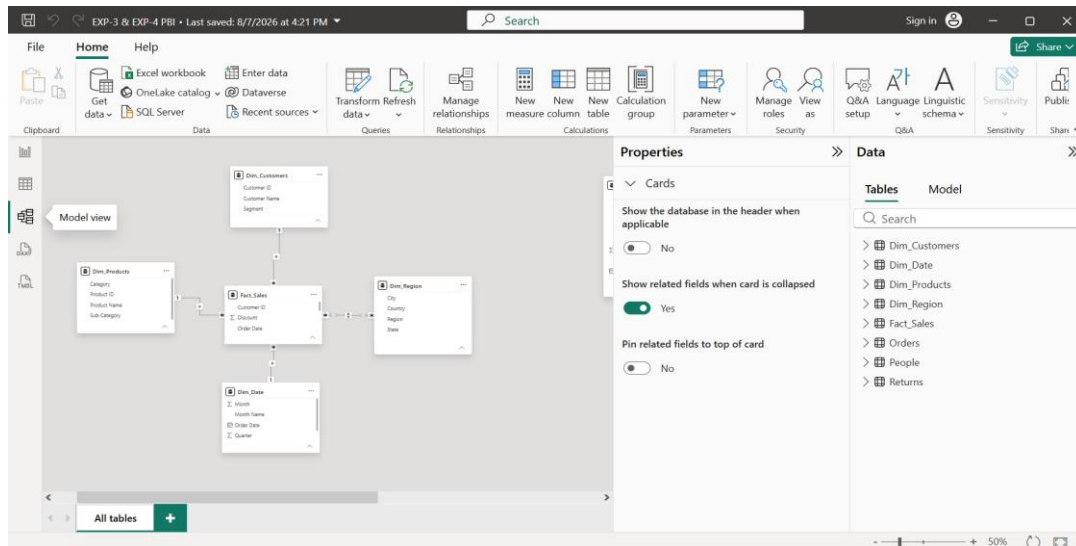
- Fact_Sales ↔ Dim_Customer
- Fact_Sales ↔ Dim_Product
- Fact_Sales ↔ Dim_Region
- Fact_Sales ↔ Dim_Date

Configure relationships.

Set the relationships appropriately as Many-to-One, with the Fact Table on the many side and the corresponding Dimension Table on the one side.

Step 18: Arrange the model.

Arrange the tables visually so that:



This forms the Star Schema with Fact_Sales at the centre and the Dimension Tables . surrounding it.

Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.

Create visualizations using the dimensional model to analyse the data.

Examples include:

- Sales by Category
- Profit by Region
- Sales Trend by Year

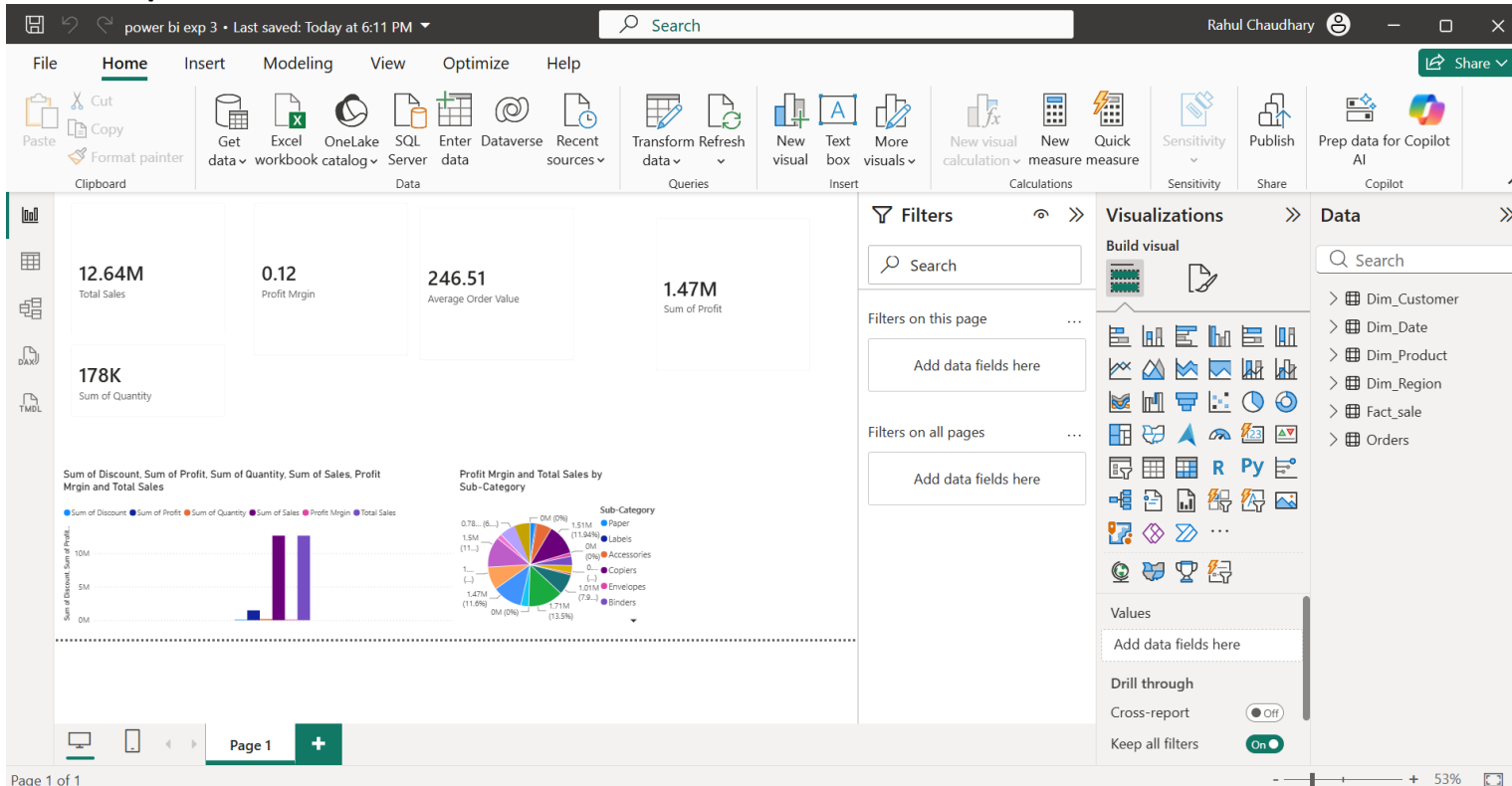
The reference specifies these as examples of visualizations for validating the model.

Step 20: Create DAX measures.

Create the following measures:

1. Total Sales = SUM(Fact_Sales[Sales])
2. Total Profit = SUM(Fact_Sales[Profit])
3. Profit Margin = DIVIDE(SUM(Fact_Sales[Profit]),SUM(Fact_Sales[Sales]))
4. Average Order Value = DIVIDE([Total Sales],DISTINCTCOUNT(Fact_Sales[Order ID]))

Step 21: Validate the Star Schema.



Use the created Dimension Tables to filter the Fact Table and verify that the relationships work correctly. Check that changes in dimensions such as Customer, Product, Region, and Date correctly affect the corresponding Fact Table data and dashboard visualizations.

6. RESULT

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

7. LEARNING OUTCOMES

- Understood the ETL process and its role in Business Intelligence.
- Learned how to extract data from a real-world business dataset using Power BI.
- Gained practical experience with Power Query Editor.
- Learned how to clean, filter, transform, and merge datasets.
- Learned how to handle duplicate records, blank values, and data type inconsistencies.
- Learned how to create relationships between multiple tables in Power BI.
- Learned how to create analytical measures using DAX.
- Understood the use of Total Sales, Total Profit, Profit Margin, and Total Orders measures.
- Learned how to create BI visualizations and dashboards for business analysis.
- Gained practical experience in preparing transformed data for reporting and decision-making.

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